

Topology optimization of structures: old and new

Grégoire Allaire

With the help of many collaborators (cited below) !

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- I - At the origin: shape optimization
- II - A review of topology optimization by density methods (homogenization + SIMP)
- III - Level set method
- IV - Heat exchangers
- V - Topology optimization of batteries
- VI - Conclusion and perspectives

I - A brief review of shape optimization

Shape optimization : a branch of optimal control theory.

Goal: minimize an **objective function** over a set of admissibles shapes Ω (including possible constraints)

$$\inf_{\Omega \in \mathcal{U}_{ad}} J(\Omega)$$

The objective function is evaluated through a partial differential equation (**state equation**)

$$J(\Omega) = \int_{\Omega} j(u_{\Omega}) dx ,$$

where u_{Ω} is the solution of

$$PDE(u_{\Omega}) = 0 \quad \text{in} \quad \Omega.$$



Example: linearized elasticity

A shape is a domain $\Omega \subset \mathbb{R}^d$ with boundary $\partial\Omega = \Gamma \cup \Gamma_N \cup \Gamma_D$.

For a load $g : \Gamma_N \rightarrow \mathbb{R}^d$, the displacement $u : \Omega \rightarrow \mathbb{R}^d$ solves

$$\begin{cases} -\operatorname{div}(A e(u)) = 0 & \text{in } \Omega \\ u = 0 & \text{on } \Gamma_D \\ (A e(u)) n = g & \text{on } \Gamma_N \\ (A e(u)) n = 0 & \text{on } \Gamma \end{cases}$$

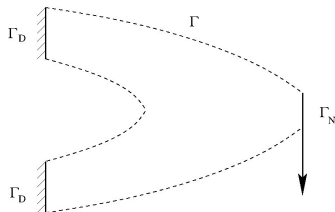
with the strain tensor $e(u) = \frac{1}{2}(\nabla u + \nabla^t u)$, and an isotropic elasticity tensor $A = 2\mu I_4 + \lambda I_2 \otimes I_2$ with $\lambda, \mu > 0$.

Typical objective function: [the compliance](#)

$$J(\Omega) = \int_{\Gamma_N} g \cdot u \, dx,$$



Admissible shapes



The set of **admissible shapes** is typically

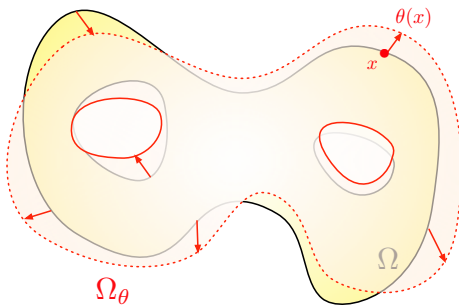
$$\mathcal{U}_{ad} = \left\{ \Omega \subset D \text{ such that } \Gamma_D \cup \Gamma_N \subset \partial\Omega \text{ and } \int_{\Omega} dx = V_0 \right\},$$

where $D \subset \mathbb{R}^d$ is given and V_0 is a prescribed volume.

Remark. Γ_D and Γ_N are fixed. **Only Γ is optimized (free boundary).**

Hadamard's method

To compute gradients and apply optimization algorithms one needs a notion of derivative for $J(\Omega)$. This is provided by [Hadamard's method](#).

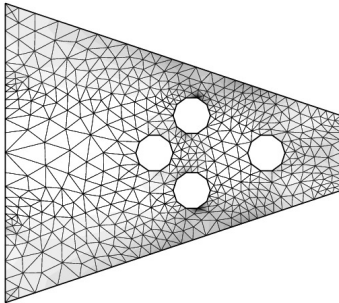


Let Ω_0 be a reference domain. Shapes are parametrized by a **vector field** θ

$$\Omega = (\text{Id} + \theta)\Omega_0 \quad \text{with} \quad \theta \in C^1(\mathbb{R}^d; \mathbb{R}^d).$$

Classical shape optimization (1)

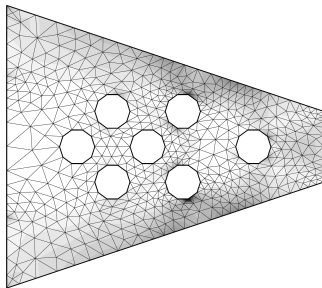
Numerical example for the cantilever:
initial shape (left), “optimal” shape (right)



- Convergence in 20 iterations.
- Global or local minimum ?
- No topology changes.

Classical shape optimization (2)

Numerical example for the cantilever:
initial shape (left), “optimal” shape (right)



- No convergence ! Rather, problem with a thin bar...
- One more local minimum !

Need for topology optimization

- It is crucial to optimize, not only the geometry, but also the **topology** of shapes.
- There are different approaches for topology optimization:
 - 1 the oldest one is the [homogenization method](#),
 - 2 SIMP (solid isotropic material with penalization),
 - 3 [level set method](#),
 - 4 phase field method,
 - 5 others (BESO, topological gradient, moving morphable components, ...).



II - Homogenization method for topology optimization

History:

- Pioneer works in the 80's: Murat-Tartar, Lurie-Cherkaev, Kohn-Strang.
- First application to structural mechanics: Bendsoe-Kikuchi (1988).
- Reference books: Allaire (2001), Bendsoe-Sigmund (2003), Cherkaev (2000), Tartar (2000).

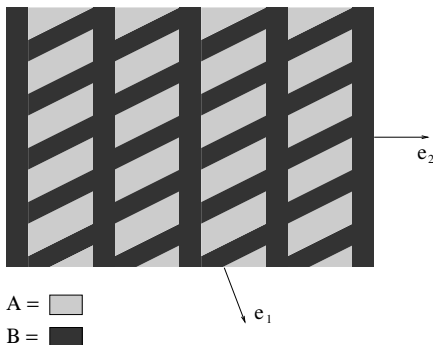
Main idea: introduce generalized admissible shapes which are **composite materials**.

- At each point there is an underlying porous microstructure.
- **New design parameters:** density $0 \leq \theta \leq 1$, and homogenized Hooke' law A^* (depending on the microstructure).
- Generalized shapes include “classical” shapes ($\theta = 0$ or 1).
- **Topology changes are allowed.**



Composite materials

- Defined rigorously by **homogenization** as the "limits" of phase mixtures.
- Large body of work on the optimization of their properties.
- A class of (multi-scale) composites with explicit formulas: **sequential laminates**.
- **Optimal composites can be found in this class.**

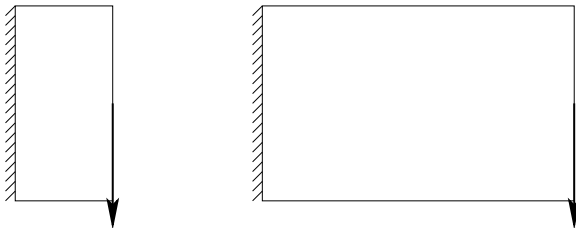


Numerical results for the homogenization theory

Goal: compute an optimal homogenized or composite design.

Algorithm for compliance minimization:

- ① Design variables: density θ , microstructure A^* .
- ② Optimal composite: rank- d sequential laminate in dimension d , with directions aligned with the stress.
- ③ Alternate minimization algorithm (optimality criteria).



Classical shapes and penalization

How to recover a classical shape from a composite design ?

Key idea: use a penalization technique which progressively forces the material density θ towards 0 or 1 values.

- 1 Compute an optimal composite design: density θ , microstructure A^* .
- 2 Continue with an additional penalization on θ .

Penalization is crucial for practical applications !

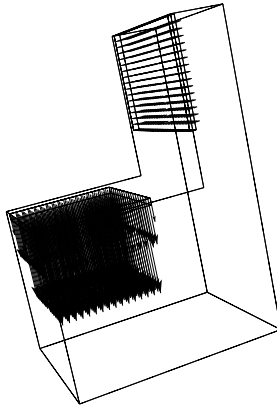


Collaborators: R. Kohn, F. Jouve, G. Francfort, E. Bonnetier.

Long cantilever with a "politically correct" initialization



3-d chair



SIMP (Solid Isotropic Material with Penalization)

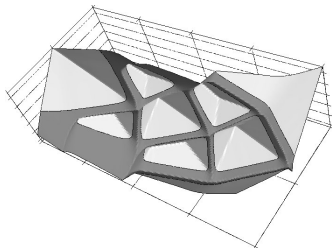
Idea. Homogenization introduces composites which are discarded at the end by penalization. *Can we simplify the approach by introducing merely a density θ ?*

- **Bendsoe:** replace the composite homogenized tensor A^* by $\theta^p A$ for some exponent $p > 1$ (for $p = 1$ this is convexification).
- This is a trick which has nothing to do with composites and homogenization.
- It works very well in practice (the difficult part is the penalization: use some kind of continuation).
- Almost all softwares are based on SIMP.
- *The homogenization method was "killed" by SIMP !*
- There is a recent resurrection of the homogenization method for lattice materials because of additive manufacturing.



III - Level set method

General method (due to Osher and Sethian) with many applications !



A shape $\Omega \subset D$ is parametrized by a **level set** function

$$\psi(x) < 0 \Leftrightarrow x \in \Omega, \quad \psi(x) > 0 \Leftrightarrow x \in (D \setminus \Omega)$$

Assume that the shape $\Omega(t)$ evolves in time t with a normal velocity $V(t, x)$. Then its motion is governed by the following **Hamilton Jacobi equation**

$$\frac{\partial \psi}{\partial t} + V|\nabla_x \psi| = 0 \quad \text{in } D.$$



Some details on the level set method

The level set algorithm allows topology changes.

The advection velocity V is deduced from the shape gradient of the objective function ([Hadamard's method](#)).

Hilbertian extension-regularization for the velocity.

Constrained optimization is taken into account by a [null space algorithm](#).

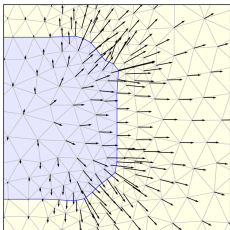
Geometrical constraints are easy to implement.

[Body-fitted shape optimization](#) is implemented by remeshing with `mmg`.

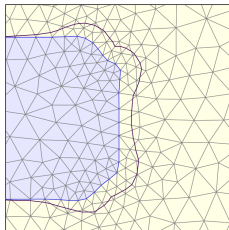
Collaborators: C. Dapogny, F. Feppon, F. Jouve.



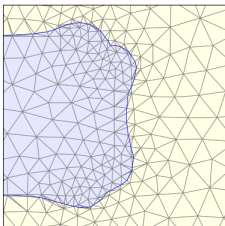
Exact remeshing with Mmg



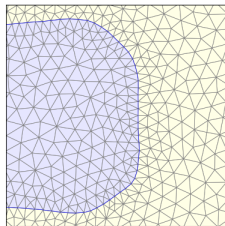
Initial interface



Zero-level set after advection

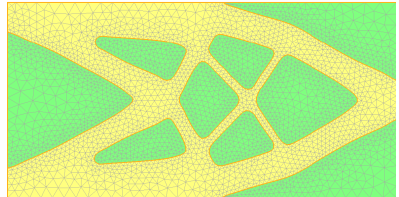
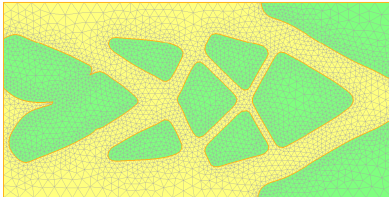
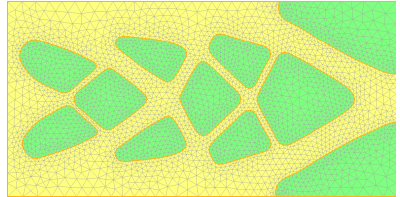
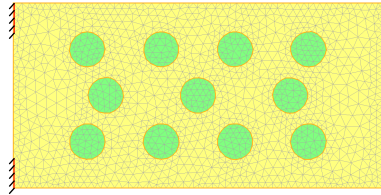


Cut mesh (bad quality)

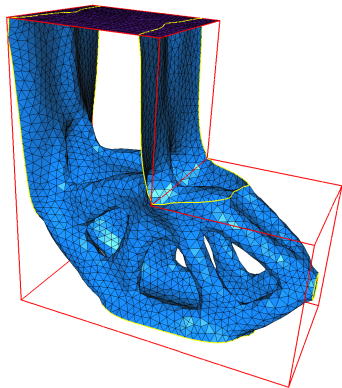


Adapted mesh

Remeshing with mmg



Compliance minimization with a weight constraint

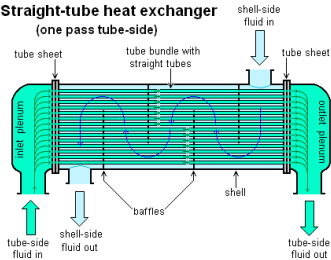


► L-beam

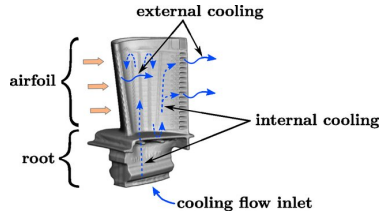
IV - Heat exchangers

F. Feppon, G. Allaire, C. Dapogny, P. Jolivet,

**Straight-tube heat exchanger
(one pass tube-side)**



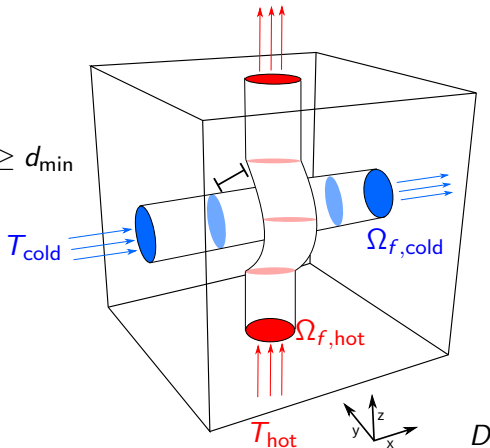
heat exchanger



turbine blade with internal cooling

Test case: 3D fluid-to-fluid heat exchanger

$$d(\Omega_{f,\text{hot}}, \Omega_{f,\text{cold}}) \geq d_{\min}$$



A non-mixing constraint for the two fluids is imposed with a minimal distance $d_{\min}$.

Physical models:

- Incompressible Navier-Stokes equations for the two fluids.
- Steady-state heat equation in the fluid (including convection) and in the surrounding solid.

Objective functions and constraints:

- Maximize the heat exchange between the two fluids.
- Constraint on the pressure drop in the two fluid channels.
- Constraint on the volumes of the fluid channels.
- Minimal distance between the hot and cold fluids to guarantee non-mixing.
- Allow for topology changes.
- Track exactly the fluid-solid interface.

Numerical values

Non-dimensional values.

Domain $D = (0, 1)^3$, inlet radius $a = 0.1$

Re	$d_{\min}$	ρ	k_f	k_s	Pe	c_p	DP ₀	V_0	T_{hot}	T_{cold}
100	0.1	10	1	10	2,000	200	3.72	0.15	100	0

Initial mesh: 3.8 million tetrahedra ($2.3 \cdot 10^6$ in the fluid domain).

Final mesh: 1.7 million tetrahedra (686,000 in the fluid domain).

Optimization: 360 iterations.



Initialization

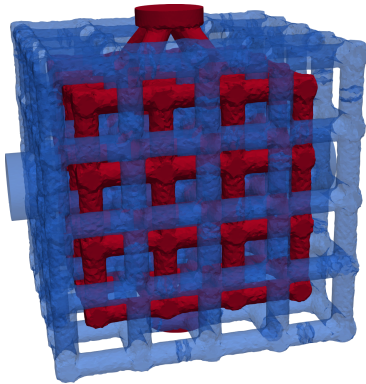


Figure: Initial distribution of fluid considered for the 3D heat exchanger. The hot and cold phase are depicted in red and blue respectively and are disjoint regions.

Optimal design (fluid parts)

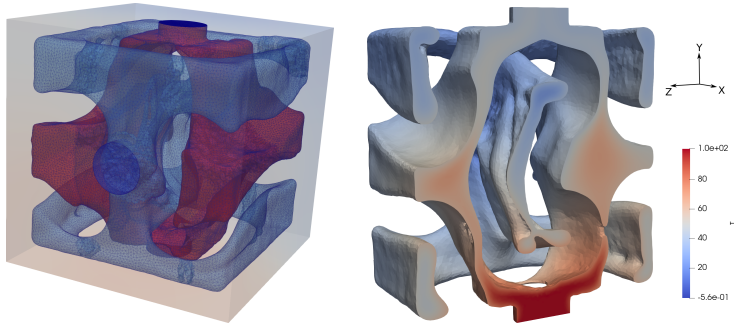


Figure: Optimized channels of the cold and hot fluids, respectively colored in blue and red (left), temperature distribution (right).

Optimal design (solid part)

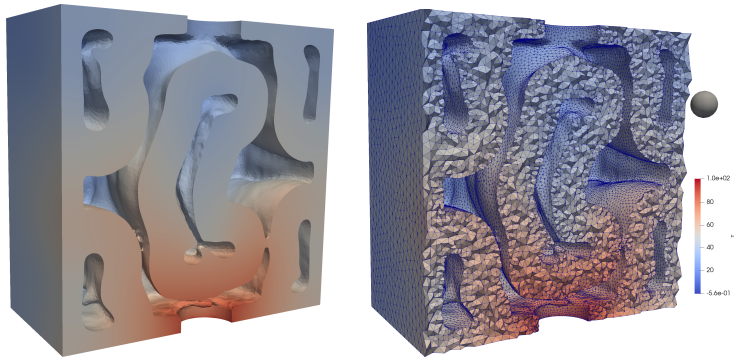
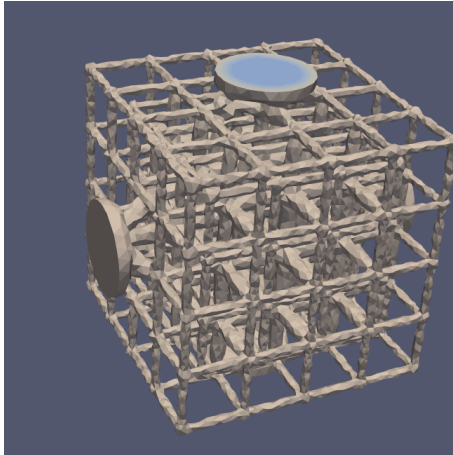
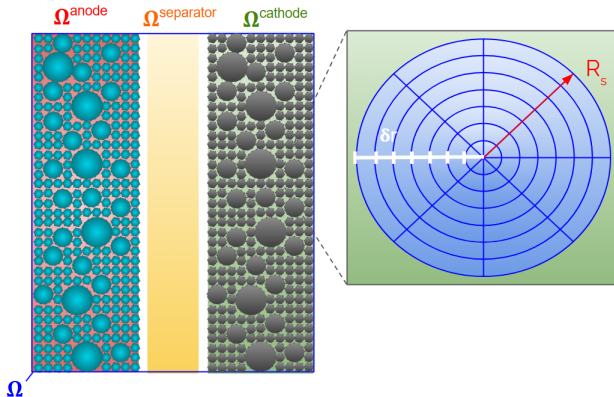


Figure: Temperature field in the solid, respecting the constraint on the minimum wall thickness (indicated by the small ball on the right).

Iteration history (including topology changes)



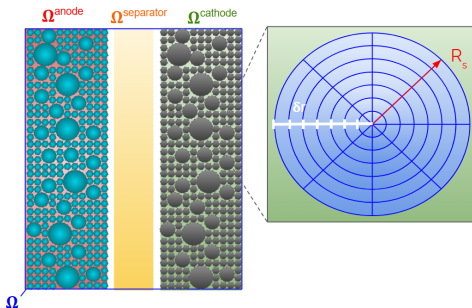
V - Topology optimization of batteries



- ☛ A battery cell is made of two porous electrodes (anode and cathode) separated by an inert separator.
- ☛ The interfaces between the separator and the electrodes, as well as the porosity, will be the **optimization variables**.

Joint work with Richard Joly and Romain de Loubens.

DFN model (Doyle, Fuller, Newman)



- ❖ The DFN model is an **homogenized and multiscale model**.
- ❖ Each electrode is a **porous medium** with spherical solid active particle (containing lithium) and a fluid electrolyte (transporting lithium ions).
- ❖ The lithium ions can diffuse through the separator.
- ❖ The vertical boundaries are **current collector** with constant electric potential.

Cell geometry (size 2.10^{-4}m):

$$\Omega = \Omega^n \cup \Omega^{sp} \cup \Omega^p$$

with $n = \text{anode}$, $sp = \text{separator}$ and $p = \text{cathode}$.

The solid active particles are balls of radius $R_s \approx 4.10^{-6}\text{m}$.

There are 4 equations for 4 unknowns:

- ☞ c_e concentration of lithium ions,
- ☞ c_s fraction of solid lithium in the active particles,
- ☞ φ_e ionic potential of the electrolyte,
- ☞ ϕ_s solid-phase electric potential.

The equation for c_s is microscopic and the other ones macroscopic.

DFN model (ctd.)

Macroscale mass conservation for the lithium ions:

$$\varepsilon_e \frac{\partial c_e}{\partial t} - \nabla \cdot (\mathbf{D}_{e,\text{eff}} \nabla c_e) = (1 - t_+^0) a_s \frac{i_{se}}{F} \quad \text{in } \Omega \times (0, T_{\max})$$

Microscale mass conservation for the solid lithium:

$$\begin{aligned} \frac{\partial c_s}{\partial t} - \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 D_s \frac{\partial c_s}{\partial r} \right) &= 0 \quad \text{in } (0, R_s) \times (0, T_{\max}), \\ -D_s \partial_r c_s(R_s) &= \frac{1}{F} i_{se} \quad \text{in } (0, T_{\max}) \end{aligned}$$

Charge conservation for the electrolyte:

$$\begin{aligned} & -\nabla \cdot (\mathbf{\Lambda}_{e,\text{eff}} \nabla \varphi_e) \\ & + \nabla \cdot \left(2\mathbf{\Lambda}_{e,\text{eff}} (1 - t_+^0) (1 + \delta_e) \frac{\nabla c_e}{c_e} \right) = a_s i_{se} \quad \text{on } \Omega \times (0, T_{\max}) \end{aligned}$$

Charge conservation for the solid electric potential:

$$-\nabla \cdot (\mathbf{\Lambda}_{s,\text{eff}} \nabla \phi_s) = -a_s i_{se} \quad \text{in } \Omega^n \cup \Omega^p \times (0, T_{\max})$$



Butler-Volmer relation

All four equations feature the non-linear (and coupling) Butler-Volmer relation.

$$\begin{aligned}i_{se} &= 2\sqrt{\left(\frac{c_e}{c_{e,\text{ref}}}\right) c_s^* (c_{s,\text{max}} - c_s^*)} \sinh(\eta) && \text{in } \Omega^n \cup \Omega^p \times (0, T_{\text{max}}), \\ \eta &= \frac{RT_0}{2F} \left[\phi_s - \varphi_e - U_o \left(\frac{c_s^*}{c_{s,\text{max}}} \right) \right] && \text{in } \Omega^n \cup \Omega^p \times (0, T_{\text{max}}), \\ c_s^* &= c_s(R_s) && \text{in } \Omega^n \cup \Omega^p \times (0, T_{\text{max}}).\end{aligned}$$

U_o is called the standard potential or open circuit voltage (OCV) and is a non-linear function given by experiments.



Shape optimization results for batteries

For fixed final time $T_{\max} = 1000s$, we maximize the objective function

$$\max_{\Omega} \mathcal{E}(\Omega, T_{\max}) = \int_0^{T_{\max}} \overline{\phi}(\Omega, t) |I_0| dt,$$

which is the **total energy** delivered by the battery.

We impose **two geometrical constraints**

- Volume constraint

$$\frac{\text{Vol}(\Omega^n)}{\text{Vol}(\Omega^p)} = \frac{\text{Vol}(\Omega_{\text{init}}^n)}{\text{Vol}(\Omega_{\text{init}}^p)}$$

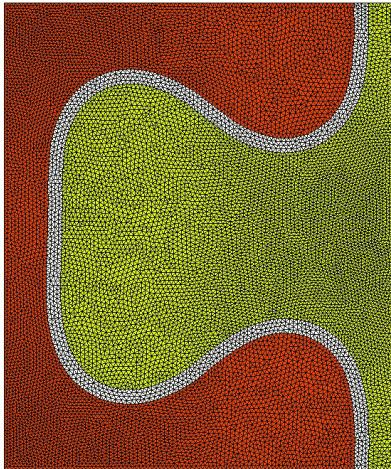
- Minimal thickness of the separator

$$d(\Gamma_{\text{sp}}^n, \Gamma_{\text{sp}}^p) \geq d_{\min}$$

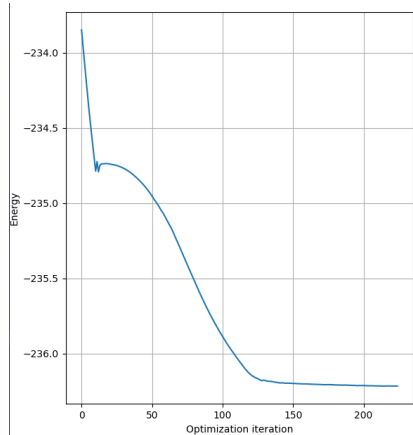


Initialization with straight interfaces $T_{\max} = 1000$

Optimization: 224



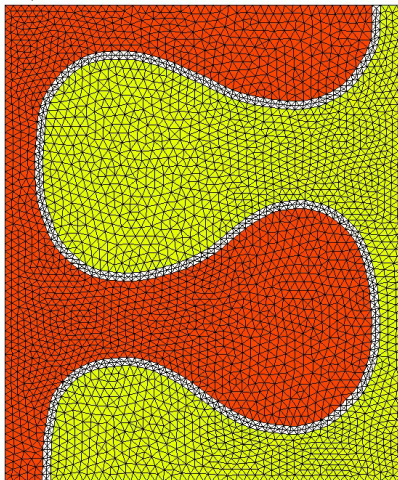
Optimized design Ω



Objective function history

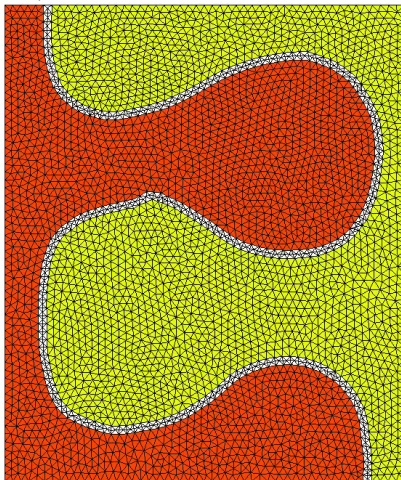
Initialization with perturbed interfaces $T_{\max} = 1000$

optim: 129



Optimized design for mode 1

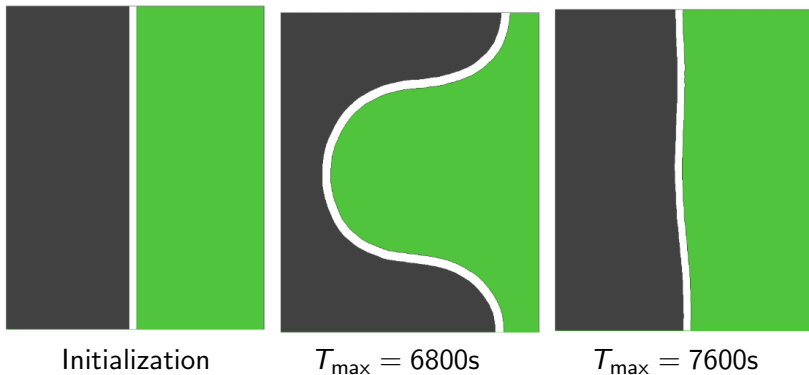
Optim Iteration : 319



Optimized design for mode 3

Influence of the final time on the design

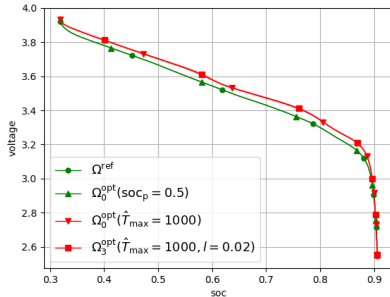
Starting from straight interfaces, we study the influence of the final time on the optimal design.



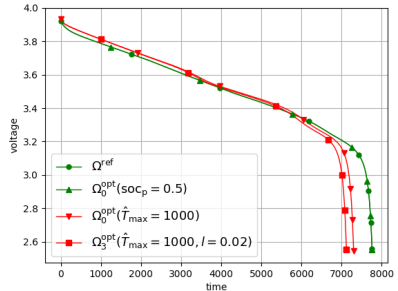
Optimizing on longer times yields less interdigitated interfaces.



Discussion on the geometric optimization experiments



Voltage versus SOC



Voltage versus time

Topology optimization of porosity

- ❖ The cell geometry is now fixed.
- ❖ The porosity within each electrode ε_e or equivalently the active material distribution $\varepsilon_s = 1 - \varepsilon_e$ are the new optimization variables.
- ❖ The porosity is constant in the separator.
- ❖ The homogenized coefficients depend on ε_s through Bruggeman's law.
- ❖ Maximization of the delivered energy $\mathcal{E}(\varepsilon_s)$.
- ❖ Same adjoints and classical parametric derivative of $\mathcal{E}(\varepsilon_s)$.



Porosity & active material distribution

Bruggeman law for homogenized coefficients

$$\mathbf{D}_{e,\text{eff}} = (1-\varepsilon_s)^\beta \mathbf{D}_{e,\text{eff}}^0, \quad \mathbf{\Lambda}_{e,\text{eff}} = (1-\varepsilon_s)^\beta \mathbf{\Lambda}_{e,\text{eff}}^0, \quad \mathbf{\Lambda}_{s,\text{eff}} = \varepsilon_s^\beta \mathbf{\Lambda}_{s,\text{eff}}^0,$$

with a constant exponent by sub-domain

$$\beta^n \approx 2.673 \quad \beta^{sp} \approx 1.837 \quad \beta^p \approx 1.855$$

The specific surface area is

$$a_s = \frac{3\varepsilon_s}{R_s}$$

Initialization of the porosity

$$\varepsilon_s^{\text{init}} = 0.7\chi_{\Omega^n} + 0.61\chi_{\Omega^{\text{sp}}} + 0.7\chi_{\Omega^p}.$$



Objective function

$$\max_{\varepsilon_s} \mathcal{E}(\varepsilon_s) := \int_0^{T_{\max}} \overline{\phi}(\varepsilon_s) |l_0| dt, \quad T_{\max} = 1000s.$$

Mass ratio constraint

$$\frac{m_{\Omega^n}}{m_{\Omega^p}} := \frac{\int_{\Omega^n} \varepsilon_s(x) dx}{\int_{\Omega^p} \varepsilon_s(x) dx} = \frac{m_{\Omega^n}^{\text{init}}}{m_{\Omega^p}^{\text{init}}}$$

Bounds constraints

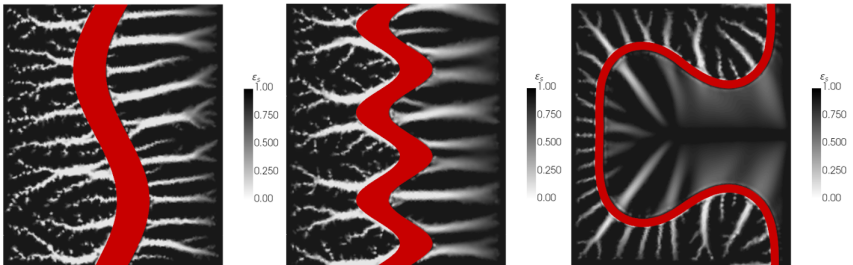
$$0 \leq \varepsilon_s^{\min} \leq \varepsilon_s \leq \varepsilon_s^{\max} \leq 1, \quad \text{in } \Omega^n \cap \Omega^p.$$

Here $\varepsilon_s^{\min} = 0.1, \varepsilon_s^{\max} = 0.9$.



First results

- ❖ Solid phase of active material is plotted in black.
- ❖ Liquid electrolyte is plotted in white.



Channels seem to locally follow the normal direction to the interfaces.

VI - Conclusion and perspectives

- Constraints induced by additive manufacturing (Martin Bihr, Benjamin Bogosel, Mathilde Boissier, Lukas Jakabcin, Matias Godoy).
- Lattice optimization by the de-homogenization method (Perle Geoffroy-Donders, Olivier Pantz).
- Electrical machine optimization (Thomas Gauthey, Maya Hage Hassan, Xavier Mininger).

